Day 139 | Unit 10 | Teacher plan

AASA Applications and Cumulative Review

Explain Data and Critique Mathematical Claims

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d, 6.EE.A.4, 6.EE.B.5, 6.EE.B.8

Learning target: I can explain Data and Critique Mathematical Claims, and explain and check my reasoning.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A useful summary names the group and attribute, units, observation count, center, variability, and shape. Explain why the chosen center and spread fit the data and mention any striking value. Use operation properties to rewrite an expression. Matching values at one input do not prove equivalence for all inputs. A single counterexample is enough to show two expressions are not equivalent. Substitute each value and decide whether the comparison is true. A continuous number-line ray represents infinitely many real-number solutions. A real situation may restrict useful solutions to whole numbers. Find the mean. Calculate each value’s distance from the mean, add those distances, then divide by the number of observations. MAD describes a typical distance from the mean in the original units.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 3, 8, 25

Day 139 | Unit 10 | Lesson

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01 / Learn

A useful summary names the group and attribute, units, observation count, center, variability, and shape. Explain why the chosen center and spread fit the data and mention any striking value. Use operation properties to rewrite an expression. Matching values at one input do not prove equivalence for all inputs. A single counterexample is enough to show two expressions are not equivalent. Substitute each value and decide whether the comparison is true. A continuous number-line ray represents infinitely many real-number solutions. A real situation may restrict useful solutions to whole numbers. Find the mean. Calculate each value’s distance from the mean, add those distances, then divide by the number of observations. MAD describes a typical distance from the mean in the original units.

Worked example 1

Reading times in minutes are 6, 7, 7, 8, 8, 9. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

1. A useful summary names the group and attribute, units, observation count, center, variability, and shape.

2. Explain why the chosen center and spread fit the data and mention any striking value.

3. Calculate the center and spread before interpreting them in context.

4. Other accurate, justified summaries are acceptable.

5. State the result in the form requested: 6 observations of reading time in minutes. Mean 15/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable..

Answer: 6 observations of reading time in minutes. Mean 15/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Worked example 2

Six students recorded reading times (minutes): 3, 8, 8, 13, 13, 18. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

1. Find the mean.

2. Mean = 63/6 = 21/2.

3. Absolute deviations: 15/2, 5/2, 5/2, 5/2, 5/2, 15/2.

4. Their sum divided by 6 is 25/6.

5. This is the average distance from the mean.

Answer: Mean 21/2 min; MAD 25/6 min

Day 139 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 4, 5. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

02 / Constructed response

Are 5(x + 3) and 5x + 15 equivalent for every x? Use a property or a counterexample.

03 / Constructed response

Which values in {9, 10, 11} satisfy x < 10? Test each by substitution.

04 / Constructed response

Reading times in minutes are 7, 10, 10, 13, 13, 16. Calculate the mean and MAD. Show all absolute deviations.

05 / Constructed response

Reading times in minutes are 5, 7, 7, 9, 9, 11. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

06 / Constructed response

Are 6(x + 2) and 6x + 12 equivalent for every x? Use a property or a counterexample.

07 / Constructed response

Which values in {10, 11, 12} satisfy x < 11? Does the real-number inequality have only these solutions? Explain.

08 / Numeric entry and interpretation

Six students recorded reading times (minutes): 3, 7, 7, 11, 11, 15. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

09 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 5, 11, 11, 17, 17, 31. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

10 / Constructed response

Are 3(x + 4) and 3x + 12 equivalent for every x? Use a property or a counterexample.

Day 139 | Unit 10 | Answer key

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1. 6 observations of reading time in minutes. Mean 7/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

2. Yes; distribution gives 5x + 15

5(x + 3) = 5x + 15. This matches.

6.EE.A.4 | Constructed response

3. 9

Values below 10 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

4. Mean 23/2 min; MAD 5/2 min

Distances: 9/2, 3/2, 3/2, 3/2, 3/2, 9/2; sum 15; divide by 6 = 5/2.

6.SP.B.5c | Constructed response

5. 6 observations of reading time in minutes. Mean 8 min, MAD 5/3 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

6. Yes; distribution gives 6x + 12

6(x + 2) = 6x + 12. This matches.

6.EE.A.4 | Constructed response

7. 10; no, infinitely many real values satisfy it

Values below 11 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

8. Mean 9 min; MAD 10/3 min

Mean = 54/6 = 9. Absolute deviations: 6, 2, 2, 2, 2, 6. Their sum divided by 6 is 10/3. This is the average distance from the mean.

6.SP.B.5c | Numeric entry and interpretation

9. 6 observations; minutes; median 14, IQR 6; mean 46/3, MAD 19/3. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 14 and half medians give IQR 6. Mean is 92/6; averaging distances gives MAD 19/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation

10. Yes; distribution gives 3x + 12

3(x + 4) = 3x + 12. This matches.

6.EE.A.4 | Constructed response